

Deployment of the FIWARE DSC

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Agenda

- FIWARE DSC deployment overview
- FIWARE DSC deployment by role
 - Provider
 - Consumer
 - Consumer & provider
 - Operator
- Recommendations and Common Issues

Doc: <https://github.com/FIWARE/data-space-connector/tree/main/doc/deployment-integration/roles>

Examples: <https://github.com/vramperez/FIWARE-DSC-deployment-examples>

Deployment overview

- TL;DR

- `helm repo add dsc https://fiware.github.io/data-space-connector/`
- `helm install dsc dsc/data-space-connector -n<ns> -f values.yaml --version 10.3.2`

- [Umbrella helm chart](#)

- Source helm chart in the [FIWARE DSC repo](#)

- Requirements:

- Kubernetes cluster
- [Helm 3](#)

Deployment by role

- Roles in a dataspace
 - **Consumer:**
 - Retrieves data or services from other participants
 - Needs the ability to issue Verifiable Credentials for its users
 - **Provider:**
 - Offers data or services to other participants
 - Needs to verify incoming credentials, enforce access policies, and host data services
 - **Consumer + Provider:**
 - Acts as both consumer and provider
 - This is the most common scenario in real-world data spaces
 - **Operator:**
 - Operates the shared trust infrastructure of the data space (Trust Anchor, Onboarding services, etc)
 - This role is typically fulfilled by a neutral governance body, not by a data space participant

Provider

■ Required components

- DID
- Authentication
 - VC Verifier
 - Credentials Config Service*
 - Trusted Issuers List
- Authorization
 - APISIX (see [deployment doc](#))
 - ODRL-PAP

■ Optional components

- VC issuer (keycloak)*
- TM Forum APIs
- Marketplace (BAE)
- Contract management
- Dataspace Protocol (FDSC-EDC and IdentityHub)

Provider: example

- Components
 - DID
 - Authentication
 - VC Verifier
 - Credentials Config Service*
 - Trusted Issuers List
 - Authorization
 - APISIX (see [deployment doc](#))
 - ODRL-PAP
 - VC issuer (keycloak)
 - TM Forum APIs
 - Marketplace (BAE)
 - Contract management
 - NGSI-LD service



Config example [here](#)

Consumer

- Required components
 - VC issuer (i.e. keycloak)
 - DID
- Optional components
 - Dataspace Protocol (FDSC-EDC + TMF APIs + IdentityHub, etc)



Config example [here](#)

Keycloak

- Realm configuration:
 - Clients
 - Client roles
 - Credential types (client scopes)
 - claims and attribute mappings
- Important things:
 - The Organization's DID private key must be the one used by Keycloak for VC signing
 - Keycloak needs a PKCS12 keystore to sign Verifiable Credentials → `initContainer`
 - Keycloak image: `quay.io/seamware/keycloak:26.6.3`
 - [Compatible wallets](#)



Config example [here](#)

Consumer + provider

- Required components
 - VC issuer (keycloak)
 - DID
 - Authentication
 - VC Verifier
 - Credentials Config Service
 - Trusted Issuers List
 - Authorization
 - APISIX
 - ODRL-PAP
- Optional components
 - TM Forum APIs
 - Marketplace (BAE)
 - Contract management



Config example [here](#)

Operator

- Required components

- Trusted Issuers Registry (trust-anchor)

TL;DR

- `helm repo add dsc https://fiware.github.io/data-space-connector/`
- `helm install trust-anchor dsc/trust-anchor -n<ns> -f values.yaml --version 1.0.0`

- Optional components:

- Central Marketplace
- Central VC issuer
- Onboarding portal



Config example [here](#)

Deployment by role

- Component matrix

Umbrella component	Sub-umbrella component	Component	Role	Consumer	Provider	Consumer + Provider	Operator
decentralized-iam	vc-authentication	VCVerifier	Validates VCs and exchanges them for tokens	-	Required	Required	-
		credentials-config-service	Holds the information which VCs are required for accessing a service	-	Required	Required	-
		trusted-issuers-list	Acts as Trusted Issuers List by providing an EBSI Trusted Issuers Registry API	-	Required	Required	-
	odrl-authorization	APISIX	APISIX as API-Gateway with an OPA plugin	-	Required	Required	-
		OPA	OpenPolicyAgent as the API Gateway's Sidecar	-	Required	Required	-
		odrl-pap	Allowing to configure ODRL policies to be used by the OPA	-	Required	Required	-
-	-	Keycloak	Issuer of VCs (OID4VCI)	Required	Required (*)	Required	-
-	-	DID	Organization's DID document	Required	Required	Required	-
-	-	tmforum-api and marketplace	Implementation of the TMForum APIs for handling contracts and FIWARE BAE	-	Optional	Optional	-
-	-	contract-management	Notification listener for contract management events out of TMForum	-	Optional	Optional	-
-	-	FDSC-EDC	Eclipse Dataspace Components (Dataspace Protocol)	Optional	Optional	Optional	-
-	-	Trusted Issuers Registry	Central registry of trusted participants	-	-	-	Required

Recommendations and Common Issues

- Prefer `did:web` over `did:key` for organizations and real envs
- The Organization's DID private key must be the one used by Keycloak for VC signing
- VC Verifier signing key must be RSA (APISIX does not support EC) [issue](#)
- APISIX ingress limitations → use FIWARE DSC `extraManifests`
- Use cert-manager for ingress TLS certificates and for internal signing certificates (i.e. verifier)
- Use the TM Forum 'all-in-one' deployment for resource-constrained environments
- Plan and design the deployment carefully to save time. Key decisions to consider:
 - VC types, roles, etc
 - Optional components
 - Self Hosted vs external DBs

Recommendations and Common Issues

- About bitnami Helm chart dependencies:
 - APISIX etcd subchart
- Do not use `generatePasswords` for PROD → secure credentials (e.g. [sealed secrets](#), [external-secrets](#), etc)
- Use wildcard certificates with caution → more than one DID in the same organization
- BAE Marketplace matrix version:
<https://github.com/FIWARE-TMForum/Business-API-Ecosystem#component-versions>

Resources

- Roles doc:
<https://github.com/FIWARE/data-space-connector/tree/main/doc/deployment-integration/roles>
- General DSC doc: <https://github.com/FIWARE/data-space-connector/tree/main/doc>
- Local FIWARE DSC deployment examples (**JUST FOR TESTING/DEV**):
<https://github.com/FIWARE/data-space-connector/tree/main/k3s>
- Production FIWARE DSC deployment examples:
<https://github.com/vramperez/FIWARE-DSC-deployment-examples>

Q&A

Thank you!

